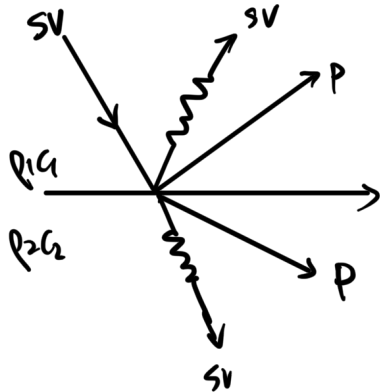


HW5

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question 1



$$\bar{\Phi}_i = A e^{-j\omega(t - p_1 x_1 - \eta_1^p x_3)}$$

$$\bar{\Phi}_r = B e^{-j\omega(t - p_1 x_1 + \eta_1^p x_3)}$$

$$\bar{\Phi}_t = C e^{-j\omega(t - p_2 x_1 + \eta_2^p x_3)}$$

$$\bar{\Phi}_t = D e^{-j\omega(t - p_2 x_1 - \eta_2^s x_3)}$$

$$\bar{\Phi}_t = E e^{-j\omega(t - p_2 x_1 - \eta_2^s x_3)}$$

$$-\omega^2 p^2 - (j\omega \eta_2)^2 + \omega^2 \eta_2^2$$

$$\vec{u} = \nabla \Phi + \nabla \times \bar{\Phi}$$

$$u_+ = \nabla \bar{\Phi}_r + \nabla \times (\bar{\Phi}_i + \bar{\Phi}_r)$$

$$= \left(\frac{\partial \bar{\Phi}_r}{\partial x_1}, 0, \frac{\partial \bar{\Phi}_r}{\partial x_3} \right) + \left(-\frac{\partial \bar{\Phi}_i}{\partial x_3} - \frac{\partial \bar{\Phi}_r}{\partial x_3}, 0, \frac{\partial \bar{\Phi}_i}{\partial x_1} + \frac{\partial \bar{\Phi}_r}{\partial x_1} \right)$$

$$= \left(\frac{\partial \bar{\Phi}_r}{\partial x_1} - \frac{\partial \bar{\Phi}_i}{\partial x_3} - \frac{\partial \bar{\Phi}_r}{\partial x_3}, 0, \frac{\partial \bar{\Phi}_r}{\partial x_3} + \frac{\partial \bar{\Phi}_i}{\partial x_1} + \frac{\partial \bar{\Phi}_r}{\partial x_1} \right)$$

$$\text{同理 } u_- = \left(\frac{\partial \bar{\Phi}_t}{\partial x_1} - \frac{\partial \bar{\Phi}_t}{\partial x_3}, 0, \frac{\partial \bar{\Phi}_t}{\partial x_3} + \frac{\partial \bar{\Phi}_t}{\partial x_1} \right)$$

$$\text{应用边界条件: } u^+ = u^-, \sigma_{31}^+ = \sigma_{31}^-, \sigma_{33}^+ = \sigma_{33}^-$$

$$(i) u_+ = u_-$$

$$\begin{cases} p_1 C - \eta_1^p A + \eta_1^p B = p_2 E - \eta_2^p D \\ -\eta_1^s C + p_1 A + p_1 B = \eta_2^s E + p_2 D \end{cases}$$

水平位移连续

垂直位移连续

$$(ii) \sigma_{31}^+ = \sigma_{31}^-$$

$$\sigma_{31} = 2\mu e_{31} = 2\mu \cdot \frac{1}{2} \left(\frac{\partial u_3}{\partial x_1} + \frac{\partial u_1}{\partial x_3} \right) = \mu \left(\frac{\partial u_3}{\partial x_1} + \frac{\partial u_1}{\partial x_3} \right)$$

$$\sigma_{31}^+ = \mu_1 \left(\frac{\partial^2 \bar{\Phi}_r}{\partial x_1 \partial x_3} + \frac{\partial^2 \bar{\Phi}_i}{\partial x_1^2} + \frac{\partial^2 \bar{\Phi}_r}{\partial x_1^2} + \frac{\partial^2 \bar{\Phi}_t}{\partial x_1 \partial x_3} - \frac{\partial^2 \bar{\Phi}_i}{\partial x_3^2} - \frac{\partial^2 \bar{\Phi}_r}{\partial x_3^2} \right)$$

$$= \mu_1 \left(2\omega^2 p_1 \eta_1^\alpha \bar{\Phi}_r|_{z=0} + \omega^2 (\eta_1^{\beta^2} - p_1^2) \bar{\Psi}_i|_{z=0} + \omega^2 (\eta_1^{\beta^2} - p_1^2) \bar{\Psi}_r|_{z=0} \right)$$

$$\sigma_{31}^- = \mu_2 \left(\frac{\partial^2 \bar{\Phi}_t}{\partial x_3 \partial x_1} + \frac{\partial^2 \bar{\Psi}_t}{\partial x_1^2} + \frac{\partial^2 \bar{\Psi}_t}{\partial x_1 \partial x_3} - \frac{\partial^2 \bar{\Psi}_t}{\partial x_3^2} \right)$$

$$= \mu_2 \left(-2\omega^2 p_2 \eta_2^\alpha \bar{\Phi}_t|_{z=0} + \omega^2 (\eta_2^{\beta^2} - p_2^2) \bar{\Psi}_t|_{z=0} \right)$$

$$\therefore \mu_1 [2p_1 \eta_1^\alpha C + (\eta_1^{\beta^2} - p_1^2)(A+B)] = \mu_2 [-2p_2 \eta_2^\alpha E^\alpha + (\eta_2^{\beta^2} - p_2^2) D] \text{ 切向应力连续}$$

$$(iii) \sigma_{33}^+ = \sigma_{33}^-$$

$$\begin{aligned}\sigma_{33} &= \lambda e_{kk} + 2\mu e_{33} = \lambda(e_{11} + e_{22} + e_{33}) + 2\mu \cdot \frac{1}{2} \left(\frac{\partial u_3}{\partial x_3} - 2 \right) \\ &= \lambda \left(\frac{\partial u_1}{\partial x_1} + \frac{\partial u_3}{\partial x_3} \right) + 2\mu \frac{\partial u_3}{\partial x_3} \\ &= \lambda \frac{\partial u_1}{\partial x_1} + (\lambda + 2\mu) \frac{\partial u_3}{\partial x_3}\end{aligned}$$

$$\sigma_{33}^+ = \lambda_1 \left(\frac{\partial^2 \bar{\Phi}_r}{\partial x_1^2} - \frac{\partial^2 \bar{\Phi}_r}{\partial x_3 \partial x_1} - \frac{\partial^2 \bar{\Phi}_r}{\partial x_3 \partial x_1} \right) + (\lambda_1 + 2\mu_1) \left(\frac{\partial^2 \bar{\Phi}_r}{\partial x_3^2} + \frac{\partial^2 \bar{\Phi}_r}{\partial x_1 \partial x_3} + \frac{\partial^2 \bar{\Phi}_r}{\partial x_1 \partial x_3} \right)$$

$$\sigma_{33}^- = \lambda_2 \left(\frac{\partial^2 \bar{\Phi}_t}{\partial x_1^2} - \frac{\partial^2 \bar{\Phi}_t}{\partial x_3 \partial x_1} \right) + (\lambda_2 + 2\mu_2) \left(\frac{\partial^2 \bar{\Phi}_t}{\partial x_3^2} + \frac{\partial^2 \bar{\Phi}_t}{\partial x_1 \partial x_3} \right)$$

$$\therefore \lambda_1 \left(-\rho^2 C + \rho_1 \eta_1^\beta A - \rho_1 \eta_1^\beta B \right) + (\lambda_1 + 2\mu_1) \left(-\eta_1^\alpha C - \rho_1 \eta_1^\beta A + \rho_1 \eta_1^\beta B \right)$$

$$= \lambda_2 \left(-\rho_2^2 E + \rho_2 \eta_2^\beta D \right) + (\lambda_2 + 2\mu_2) \left(-\eta_2^\alpha E - \rho_2 \eta_2^\beta D \right) \quad \text{反何应力连续}$$

$$-\lambda_1 (\rho_1^2 + \eta_1^{\alpha^2}) - 2\mu_1 \eta_1^{\alpha^2} C$$

$$= -\frac{\lambda_1}{\alpha_1^2} - 2\mu_1 \eta_1^{\alpha^2} C = -\mu_1 (\eta_1^{\beta^2} - 2\eta_1^{\alpha^2} - \rho^2 + 2\eta_1^{\alpha^2}) C = -\mu_1 (\eta_1^{\beta^2} - \rho^2) C$$

$$\begin{aligned}\lambda_1 + 2\mu_1 &= \rho \alpha^2 \cdot \\ \mu_1 &= \rho \beta^2\end{aligned} \quad \frac{\lambda_1}{\alpha^2} + 2\frac{\mu_1}{\alpha^2} = \rho = \frac{\mu_1}{\beta^2} \quad \begin{aligned}\frac{\lambda_1}{\alpha^2} &= \mu_1 (\eta_1^{\beta^2} + \rho^2 - 2\eta_1^{\alpha^2} - \rho^2) \\ &= \mu_1 (\eta_1^{\beta^2} - \eta_1^{\alpha^2} - \rho^2)\end{aligned}$$

$$\therefore 2\mu_1 \rho \eta_1^\beta B - \mu_1 (\eta_1^{\beta^2} - \rho^2) C + 2\mu_2 \rho \eta_2^\beta D + \mu_2 (\eta_2^{\beta^2} - \rho^2) E = -2\mu_1 \rho \eta_1^\beta A$$

$$\begin{bmatrix} \eta_1^\beta & \rho & \eta_2^\beta & -\rho \\ \rho & -\eta_1^\alpha & -\rho & \eta_2^\alpha \\ \mu_1 \mu_1 & 2\mu_1 \eta_1^\alpha \rho & -\mu_2 \mu_2 & 2\mu_2 \eta_2^\alpha \rho \\ 2\mu_1 \rho \eta_1^\beta & -\mu_1 \mu_1 & 2\mu_2 \rho \eta_2^\beta & \mu_2 \mu_2 \end{bmatrix} \begin{bmatrix} B \\ C \\ D \\ E \end{bmatrix} = \begin{bmatrix} \eta_1^\beta \\ -\rho \\ -\mu_1 \mu_1 \\ 2\mu_1 \rho \eta_1^\beta \end{bmatrix} A$$

$$|u_i| = k_{\beta_1} A$$

$$|u_{st}| = k_{\beta_2} D$$

$$R_{ss} = \frac{|u_{sr}|}{|u_i|} = \frac{B}{A}$$

$$|u_{sr}| = k_{\beta_1} B$$

$$|u_{pt}| = k_{\alpha_2} E$$

$$R_{ps} = \frac{k_{\alpha_1}}{k_{\beta_1}} \frac{C}{A} = \frac{\beta_1}{\alpha_1} \frac{C}{A}$$

$$|u_{pr}| = k_{\alpha_1} C$$

$$T_{ss} = \frac{k_{\beta_2}}{k_{\beta_1}} \frac{D}{A} = \frac{\beta_1}{\beta_2} \frac{D}{A}$$

$$T_{sp} = \frac{k_{\alpha_2}}{k_{\beta_1}} \frac{E}{A} = \frac{\beta_1}{\alpha_2} \frac{E}{A}$$

实际计算中，只需求解矩阵，然后将比值带入反射/透射系数。

question2

$$(iii) \sigma_{33}^+ = \sigma_{33}^-$$

$$\begin{aligned}\sigma_{33} &= \lambda e_{kk} + 2\mu e_{33} = \lambda(e_{11} + e_{22} + e_{33}) + 2\mu \cdot \frac{1}{2} \left(\frac{\partial u_3}{\partial x_3} - 2 \right) \\ &= \lambda \left(\frac{\partial u_1}{\partial x_1} + \frac{\partial u_3}{\partial x_3} \right) + 2\mu \frac{\partial u_3}{\partial x_3} \\ &= \lambda \frac{\partial u_1}{\partial x_1} + (\lambda + 2\mu) \frac{\partial u_3}{\partial x_3}\end{aligned}$$

$$\sigma_{33}^+ = \lambda_1 \left(\frac{\partial^2 \bar{\Phi}_r}{\partial x_1^2} - \frac{\partial^2 \bar{\Phi}_r}{\partial x_3 \partial x_1} - \frac{\partial^2 \bar{\Phi}_r}{\partial x_3 \partial x_1} \right) + (\lambda_1 + 2\mu_1) \left(\frac{\partial^2 \bar{\Phi}_r}{\partial x_3^2} + \frac{\partial^2 \bar{\Phi}_r}{\partial x_1 \partial x_3} + \frac{\partial^2 \bar{\Phi}_r}{\partial x_1 \partial x_3} \right)$$

$$\sigma_{33}^- = \lambda_2 \left(\frac{\partial^2 \bar{\Phi}_t}{\partial x_1^2} - \frac{\partial^2 \bar{\Phi}_t}{\partial x_3 \partial x_1} \right) + (\lambda_2 + 2\mu_2) \left(\frac{\partial^2 \bar{\Phi}_t}{\partial x_3^2} + \frac{\partial^2 \bar{\Phi}_t}{\partial x_1 \partial x_3} \right)$$

$$\therefore \lambda_1 \left(-\rho^2 C + \rho_1 \eta_1^\beta A - \rho_1 \eta_1^\beta B \right) + (\lambda_1 + 2\mu_1) \left(-\eta_1^\alpha C - \rho_1 \eta_1^\beta A + \rho_1 \eta_1^\beta B \right)$$

$$= \lambda_2 \left(-\rho_2^2 E + \rho_2 \eta_2^\beta D \right) + (\lambda_2 + 2\mu_2) \left(-\eta_2^\alpha E - \rho_2 \eta_2^\beta D \right) \quad \text{反何应力连续}$$

$$-\lambda_1(\rho_1^2 + \eta_1^{\alpha^2}) - 2\mu_1 \eta_1^{\alpha^2} C$$

$$= -\frac{\lambda_1}{\alpha_1^2} - 2\mu_1 \eta_1^{\alpha^2} C = -\mu_1 (\eta_1^{\beta^2} - 2\eta_1^{\alpha^2} - \rho^2 + 2\eta_1^{\alpha^2}) C = -\mu_1 (\eta_1^{\beta^2} - \rho^2) C$$

$$\begin{aligned}\lambda_1 + 2\mu_1 &= \rho \alpha^2 \cdot \\ \mu_1 &= \rho \beta^2\end{aligned} \quad \frac{\lambda_1}{\alpha^2} + 2\frac{\mu_1}{\alpha^2} = \rho = \frac{\mu_1}{\beta^2} \quad \begin{aligned}\frac{\lambda_1}{\alpha^2} &= \mu_1 (\eta_1^{\beta^2} + \rho^2 - 2\eta_1^{\alpha^2} - \rho^2) \\ &= \mu_1 (\eta_1^{\beta^2} - \eta_1^{\alpha^2} - \rho^2)\end{aligned}$$

$$\therefore 2\mu_1 \rho \eta_1^\beta B - \mu_1 (\eta_1^{\beta^2} - \rho^2) C + 2\mu_2 \rho \eta_2^\beta D + \mu_2 (\eta_2^{\beta^2} - \rho^2) E = -2\mu_1 \rho \eta_1^\beta A$$

$$\begin{bmatrix} \eta_1^\beta & \rho & \eta_2^\beta & -\rho \\ \rho & -\eta_1^\alpha & -\rho & \eta_2^\alpha \\ \mu_1 \mu_1 & 2\mu_1 \eta_1^\alpha \rho & -\mu_2 \mu_2 & 2\mu_2 \eta_2^\alpha \rho \\ 2\mu_1 \rho \eta_1^\beta & -\mu_1 \mu_1 & 2\mu_2 \rho \eta_2^\beta & \mu_2 \mu_2 \end{bmatrix} \begin{bmatrix} B \\ C \\ D \\ E \end{bmatrix} = \begin{bmatrix} \eta_1^\beta \\ -\rho \\ -\mu_1 \mu_1 \\ 2\mu_1 \rho \eta_1^\beta \end{bmatrix} A$$

$$|u_i| = k_{\beta_1} A$$

$$|u_{st}| = k_{\beta_2} D$$

$$R_{ss} = \frac{|u_{sr}|}{|u_i|} = \frac{B}{A}$$

$$|u_{sr}| = k_{\beta_1} B$$

$$|u_{pt}| = k_{\alpha_2} E$$

$$R_{ps} = \frac{k_{\alpha_1}}{k_{\beta_1}} \frac{C}{A} = \frac{\beta_1}{\alpha_1} \frac{C}{A}$$

$$|u_{pr}| = k_{\alpha_1} C$$

$$T_{ss} = \frac{k_{\beta_2}}{k_{\beta_1}} \frac{D}{A} = \frac{\beta_1}{\beta_2} \frac{D}{A}$$

$$T_{sp} = \frac{k_{\alpha_2}}{k_{\beta_1}} \frac{E}{A} = \frac{\beta_1}{\alpha_2} \frac{E}{A}$$

$$\therefore \begin{bmatrix} \eta_1^\alpha & \rho & \eta_2^\alpha \\ 2\mu_1 p \eta_1^\alpha & \mu_1 M_1 & 0 \\ -\mu_1 M_1 & 2\mu_1 p \eta_1^\beta & \rho_2 \end{bmatrix} \begin{bmatrix} R_{PP} \\ R_{PS} \\ T_{PP} \end{bmatrix} = \begin{bmatrix} \eta_1^\alpha \\ 2\mu_1 p \eta_1^\alpha \\ \mu_1 M_1 \end{bmatrix}$$

$$|u_i| = k\alpha_1 A$$

$$|u_{pr}| = k\alpha_1 B$$

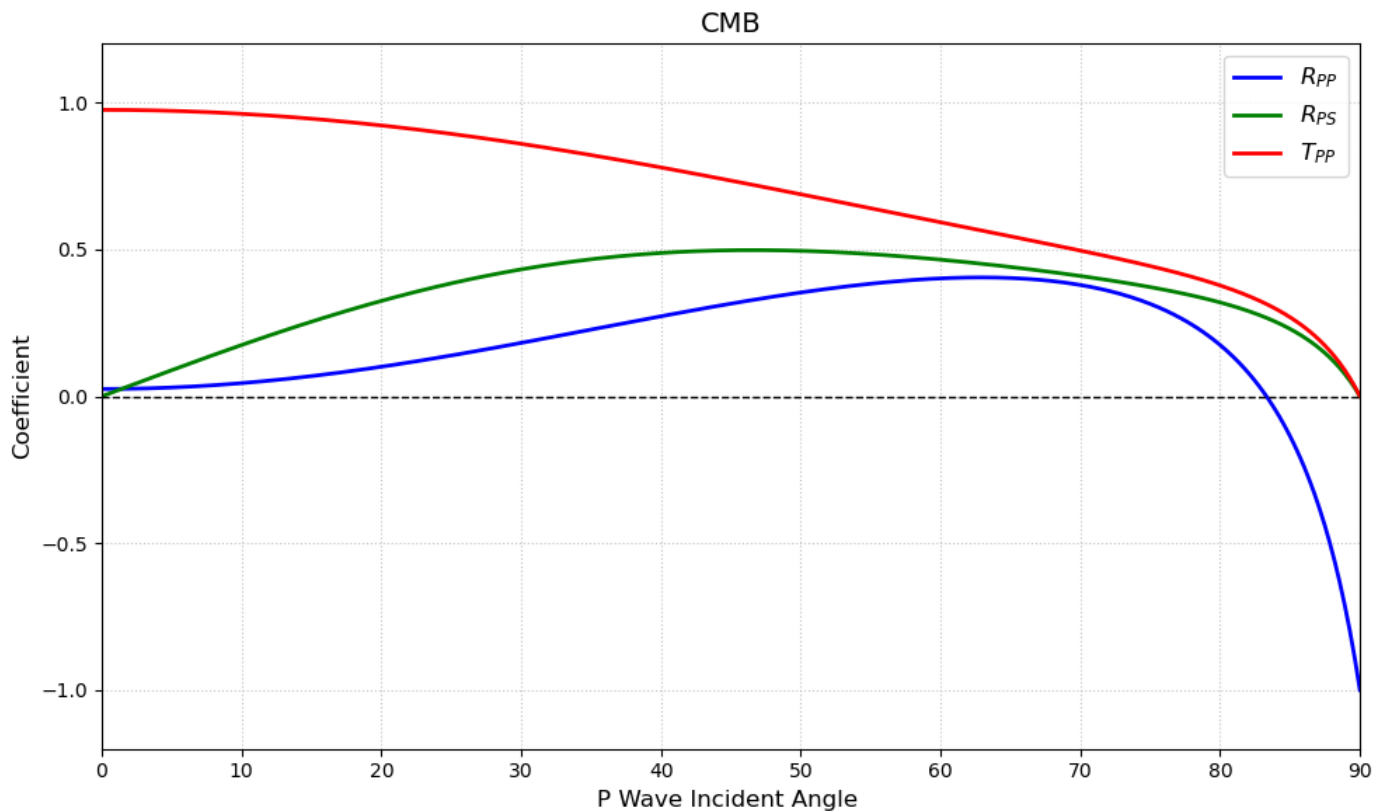
$$|u_{sr}| = k\beta_1 C$$

$$|u_{pt}| = k\alpha_2 D$$

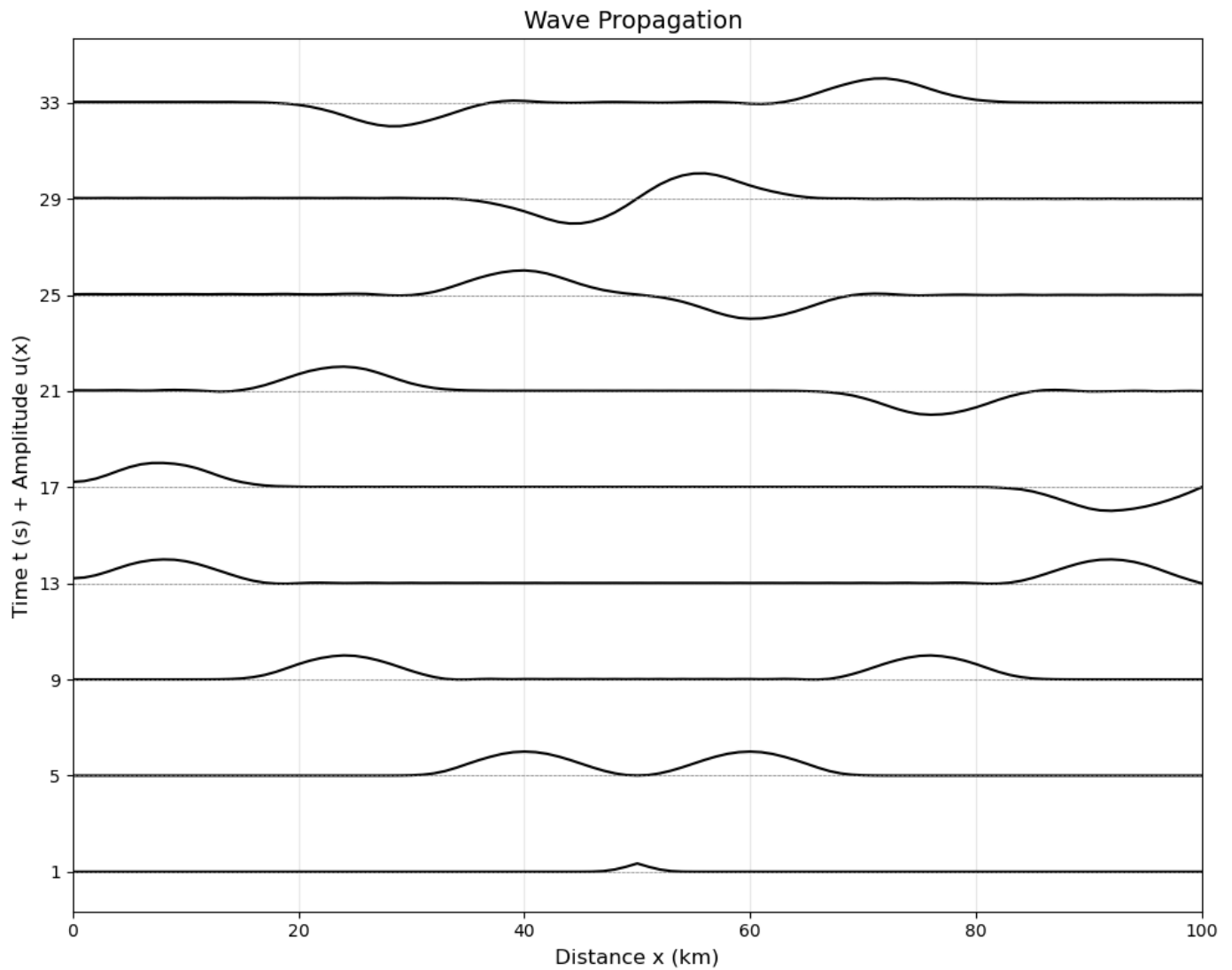
$$R_{PP} = \frac{B}{A}$$

$$R_{PS} = \frac{\alpha_1}{\beta_1} \frac{C}{A}$$

$$T_{PP} = \frac{\alpha_1}{\alpha_2} \frac{D}{A}$$



question3



- **验证波速：**震源作用时间为 5 s。在 $t = 5$ s 时，波刚脱离震源，波峰大约位于 $x = 40$ km 和 $x = 60$ km 处。在 $t = 9$ s 时，左侧波峰移动到了 $x = 24$ km 处。
- 波峰移动距离： $\Delta x = 40 - 24 = 16$ km。
- 时间差： $\Delta t = 9 - 5 = 4$ s。
- 计算波速： $v = 16/4 = 4$ km/s，完美吻合题目给定的 β 值。

- **左端 (0 km 自由边界)：**波发生全反射时，**相位保持不变**。原本是正向的波峰（向上的位移），反射回来后依然是正向的波峰。这从 $t = 13$ s 到 $t = 17$ s 左侧端点的波形演化可以清晰看到。
- **右端 (100 km 固定边界)：**波发生全反射时，位移被强制为 0，导致**相位反转**。原本是正向的波峰，反射回来后变成了负向的波谷（向下的位移）。这在 $t = 13$ s 到 $t = 17$ s 时的右侧端点表现得非常明显。

- **波的叠加与穿透：**当两列反射波在中心相遇时，它们遵循**线性叠加原理**。由于左侧反射波是正波，右侧反射波是负波，当它们在大约 $t = 25$ s 于 $x = 50$ km 处相遇时，会发生**相消干涉**，使得该区域的总位移几乎为

零。交汇过后，它们互不影响地继续向原本的方向传播，保持各自的形状和速度。